

SIMATS Engineering

SIMATS Online Programming Lab — Python Programming (Class Test)

Course Name	Python Programming
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Topic	CO3-AT2-CLASS TEST
Question	Q1 — List Operations on Temperature Data (1 Mark)

Q1. [1 Mark]

Write Python statements to perform the following on the list `temp = [30, 32, 31, 29, 35, 33, 30]`:

- (a) Append the temperature 34.
- (b) Remove temperature 29.
- (c) Sort the list in descending order.
- (d) Display the sorted list using `print()`.

Approach / Design

- **Appending:** `list.append(34)` adds the new temperature 34 to the end of the list.
- **Removing:** `list.remove(29)` deletes the first occurrence of the value 29 from the list.
- **Sorting:** `list.sort(reverse=True)` sorts the list in place in descending order.
- **Displaying:** `print(temp)` outputs the final sorted list.

Python Program

```
"""
CO3-AT2 - Class Test
Q1: List Operations on Temperature Data [1 Mark]
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Performs the following operations on temp = [30, 32, 31, 29, 35, 33, 30]:
(a) Append the temperature 34.
(b) Remove temperature 29.
(c) Sort the list in descending order.
(d) Display the sorted list using print().
"""

temp = [30, 32, 31, 29, 35, 33, 30]

# (a) Append the temperature 34
temp.append(34)

# (b) Remove temperature 29
temp.remove(29)

# (c) Sort the list in descending order
temp.sort(reverse=True)

# (d) Display the sorted list
print(temp)
```

Sample Output

[35, 34, 33, 32, 31, 30, 30]

Conclusion

Starting from temp = [30, 32, 31, 29, 35, 33, 30], appending 34 and removing 29 gives [30, 32, 31, 35, 33, 30, 34]. Sorting this list in descending order using sort(reverse=True) produces the final list [35, 34, 33, 32, 31, 30, 30], which matches the expected output when printed.